



THE LIVING AGE

BUSINESS PLAN • CLOKEY PRODUCTIONS

01 — MISSION

Build the tools that let humanity design with living systems.

The Living Age is a unified philosophy-and-technology project proving that life organizes better than control systems—and building the tools to act on that truth.

Every living system on Earth—from gene regulatory networks to forests to immune systems—self-organizes through distributed intelligence, local signals, and emergent order. No master gene runs your body. No forester runs a forest. Yet these systems are resilient, adaptive, and beautiful.

Industrial civilization does the opposite: centralized control, top-down management, engineered rigidity. The result is brittleness, constant maintenance, and environmental destruction.

The Living Age asks: what happens when you stop controlling life and start working with it?

02 — THE TWO WINGS

Media plants the seed. Technology grows the tree.

WING 1 — MEDIA (THE WHY)

Clokey Productions

YouTube channel with five interlocking series: True Republic, Sagent Creed, Media Mythology, Eight Pillars, and Bridge Episodes. Philosophical writings, the Living Book Trilogy, and a portfolio website. Media creates the audience that becomes customers.

WING 2 — TECHNOLOGY (THE HOW)

Living Works Platform

AI-driven platform for designing biological systems, organisms, and living materials. LivingWorks Engine takes English descriptions and outputs designed DNA sequences. 3D Morphogenesis Engine simulates biological growth. Technology fulfills the demand that media creates.

Media to show why. Technology to show how.

Neither is complete without the other.

03 — THE PROBLEM

Civilization still builds with blocks.

Modern construction, manufacturing, and infrastructure are based on industrial-age engineering—assemble rigid components, consume massive energy, replace when broken. The \$10 trillion global construction industry builds dead things that decay.

Meanwhile, living systems self-assemble, self-repair, adapt, and grow without centralized control. Our current tools only allow us to *analyze* biology. We have no design interface for working with life as a creative medium.

That is the gap Living Works fills.

04 — MARKET OPPORTUNITY

\$45B+ addressable across four converging industries.

MARKET	SIZE	OPPORTUNITY
Synthetic Biology	\$20B+	AI-driven organism engineering for pharmaceuticals, agriculture, and manufacturing
AI Drug Design	\$10B+	Generative models for proteins and therapeutic molecules
Bioinformatics Software	\$15B+	Enterprise tools managing biological research data and pipelines
Programmable Living Materials	Emerging	Bio-grown buildings, carbon-absorbing cities, self-repairing materials—no dominant design platform yet

Living Works is positioned as the first mover in living materials design—a market that does not yet have a standard platform.

05 — TECHNOLOGY STACK

Four core technologies building a new design paradigm.

- ▶ **Morphogenesis Engine.** 3D simulation of biological growth: branching systems, nutrient diffusion, biomechanical forces, cellular rules. The form emerges from the simulation, not from a blueprint.
- ▶ **Natural Language Bio Design.** Describe a biological structure in plain English. The system generates growth algorithms, gene pathway suggestions, and structural simulations. Powered by Claude + ESM-3.
- ▶ **BioCAD Environment.** Unified workspace for organism blueprinting, gene network visualization, tissue architecture modeling, and ecosystem design.
- ▶ **Living Materials Library.** Curated database of biological building blocks—mycelium, plant scaffolds, bacterial cellulose, coral mineralization—each with encoded growth rules.
- ▶ **LivingWorks Engine (Built).** Retrieves real genes from 220+ genomes (UCSC, NCBI). Checks cross-species compatibility (codon bias, regulatory signals, pathway conflicts). Assembles optimized constructs with insulator elements. Outputs synthesis-ready FASTA, pathway maps, and modification logs.

06 — THE NORTH STAR

Living Houses.

Not a single organism—a chimeric living system where multiple organisms cooperate:

COMPONENT	BIOLOGICAL SOURCE	FUNCTION
Load-bearing structure	Mycelium	Structural network that grows into form
Hard surfaces	Coral-derived mineralization	Calcium carbonate deposition from environment
Flexible membranes	Bacterial cellulose	Self-assembling thin films and barriers
Tensile reinforcement	Spider silk proteins	Stronger than steel by weight
Directional growth	Phototropism circuits	Light-responsive orientation
Self-repair	Planaria pathways	Regeneration from damage
Stress signaling	Bioluminescence	Visual indicator of structural health

Communities growing their own shelter from organisms that restore the ecosystems they stand in.

07 — REVENUE MODEL

Three streams: SaaS, services, IP licensing.

STREAM	TARGET CUSTOMERS	PRICING
Academic SaaS	Universities, synthetic biology labs	\$50-\$200/mo
Startup SaaS	Biotech companies, materials startups	\$500/mo
Enterprise SaaS	Architecture firms, pharmaceutical companies	\$2,000+/mo
AI Design Services	Materials companies, construction firms	Project-based
IP Licensing	Growth algorithm patents, organism systems	Recurring royalties
Media & Merch	Audience, supporters, movement builders	Direct sales

The design software funds the research. The media creates the audience that becomes customers. The moonshot delivers living structures.

08 — ROADMAP

From prototype to ecosystem.FOUND-
ATION
BUILT

Genome retrieval, compatibility engine, assembly. LivingWorks Engine retrieves real genes from 220+ genomes, checks cross-species compatibility, and outputs synthesis-ready constructs.

PHASE 1
NEXT

English-to-Protein Bridge. Natural language interface powered by Claude + ESM-3. Describe what you want in plain English, receive designed protein sequences. Capstone project.

PHASE 2
PLANNED

Regulatory prediction + visual interface. Enformer integration for predicting gene expression from sequence. First version of the BioCAD workspace.

PHASE 3
PLANNED

Gene regulatory network design. Design complete gene circuits: sensors, logic gates, and output pathways. Move from individual genes to systems.

PHASE 4
PLANNED

Full morphogenesis engine. 3D growth simulation with nutrient diffusion, biomechanical forces, and cellular rules. Design living forms that emerge from biological principles.

PHASE 5
PLANNED

Living Architecture Design Studio. Web platform combining all tools into a unified design environment. Launch SaaS tiers. Partner with architecture firms and materials companies.

PHASE 6
PLANNED

Physical living prototypes. Lab validation. First biological structures grown from designed organisms. Proof that the software produces real, living results.

PHASE 7
DESTINATION

Living houses. Communities growing their own shelter from organisms that restore ecosystems. The platform becomes the standard design environment for biological systems across industries.

09 — THE CULTURAL ENGINE

Technology and philosophy reinforce each other.

A person who watches the channel long enough should eventually ask: *"Okay—I understand this. How do I live this way?"* The technology is the answer.

A person who encounters the technology should eventually ask: *"Why does this exist? Why design with life instead of against it?"* The media is the answer.

MEDIA ENGINE

The Sagent Creed videos, True Republic series, Media Mythology episodes, and Eight Pillars series create cultural demand for biological design—the same way "Think Different" created demand for personal computers before people understood why they needed them.

TECHNOLOGY ENGINE

The platform enables biological design across scales—from genome to architecture. SaaS subscriptions, enterprise services, and IP licensing create revenue. Real-world solutions: living buildings, carbon-negative materials, programmable ecosystems.

10 — THE UNITY

Every piece points at the same truth from a different angle.

SCALE	MEDIUM	WHAT IT SHOWS
Molecular	Code	DNA is decentralized code
Cellular	Code	Cells coordinate without a master
Ecological	Philosophy	Forests organize without foresters
Political	Philosophy	True Republic > Empire
Spiritual	Media	Every tradition points to the living order
Narrative	Media	Stories encode the pattern
Personal	Story	The Sagent's path through generations
Physical	Technology	Living houses—architecture as organism

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11 — FOUNDER

Sage Clokey — Spiral Steward

Bioinformatics graduate (UCSC). Third-generation Clokey—grandson of Art Clokey, creator of Gumby. The inheritance is not the character but the ability to create stories and tools that awaken imagination. Clay to code. Story to system.

Clokey Productions is the media wing. Living Works is the technology wing. Together they form the Living Age.

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Industrial civilization built with blocks.
The next civilization will grow with living systems.